In [22]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.express as px
import statistics as st
import plotly.io as pio
pio.renderers.default = "notebook"
```

In [23]:
```
fl = pd.read_csv('Main.csv')
```

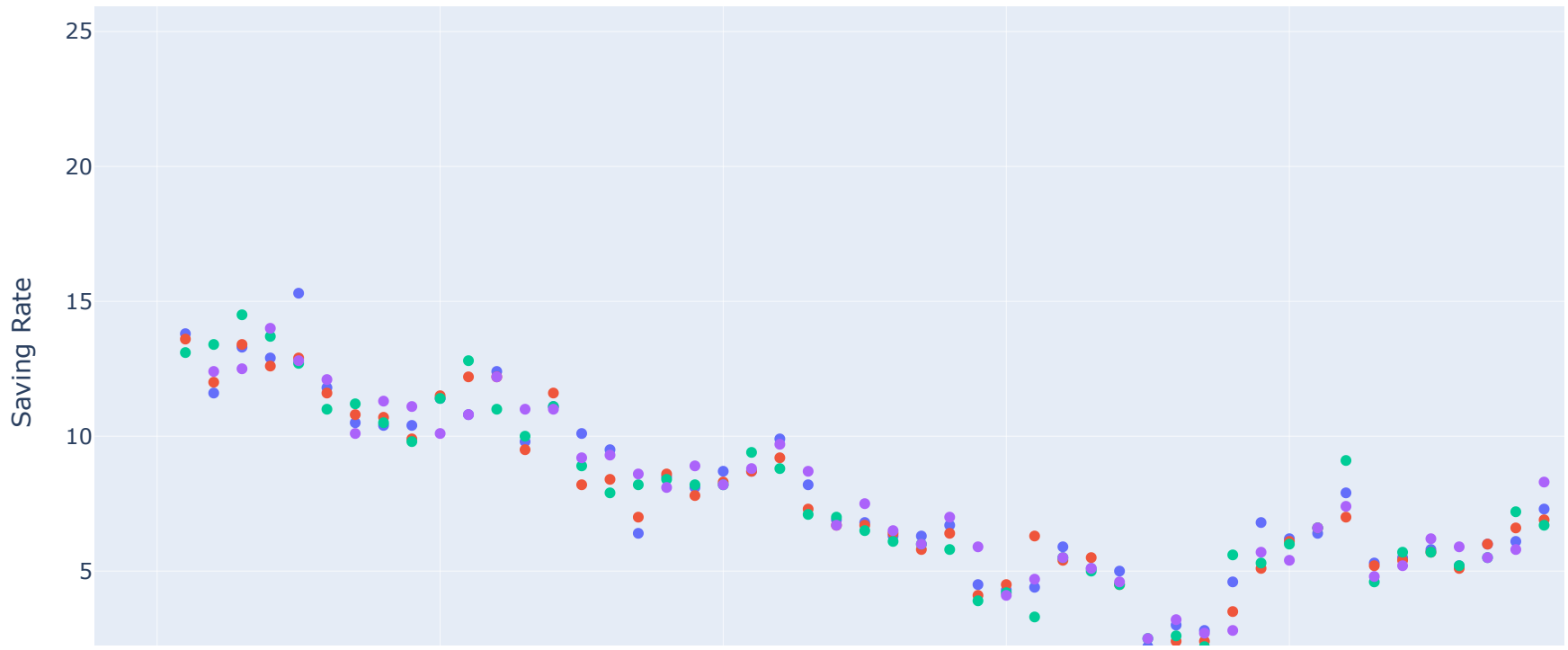
In [24]:
```
fl.head()
```

Out[24]:

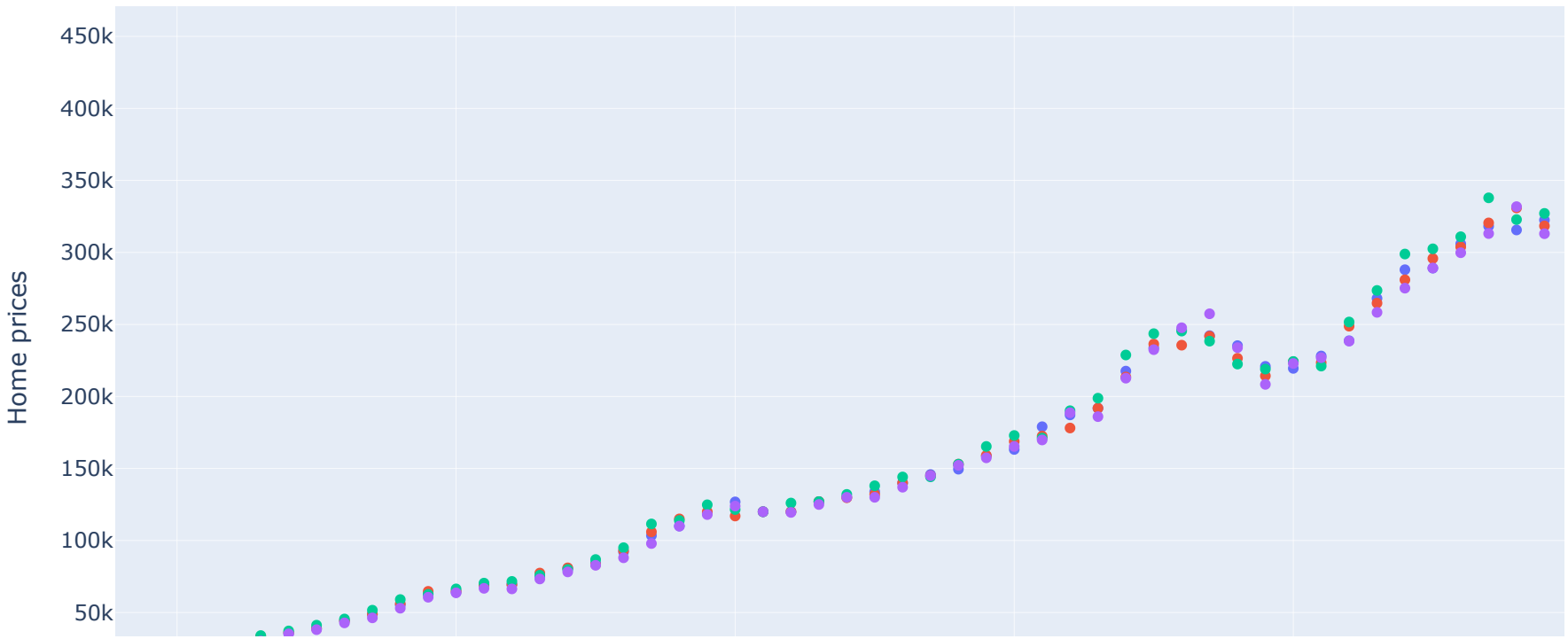
	MORTGAGE	Home prices	Expenditure	Saving Rate	Quarter	Year
0	7.41	25800	693.2	13.8	Second	1971
1	7.66	25300	705.6	13.6	Third	1971
2	7.55	25500	721.7	13.1	Fourth	1971
3	7.35	26200	739.0	12.4	First	1972
4	7.35	26800	757.4	11.6	Second	1972

First we need to observe the trends with each column of data across the years.

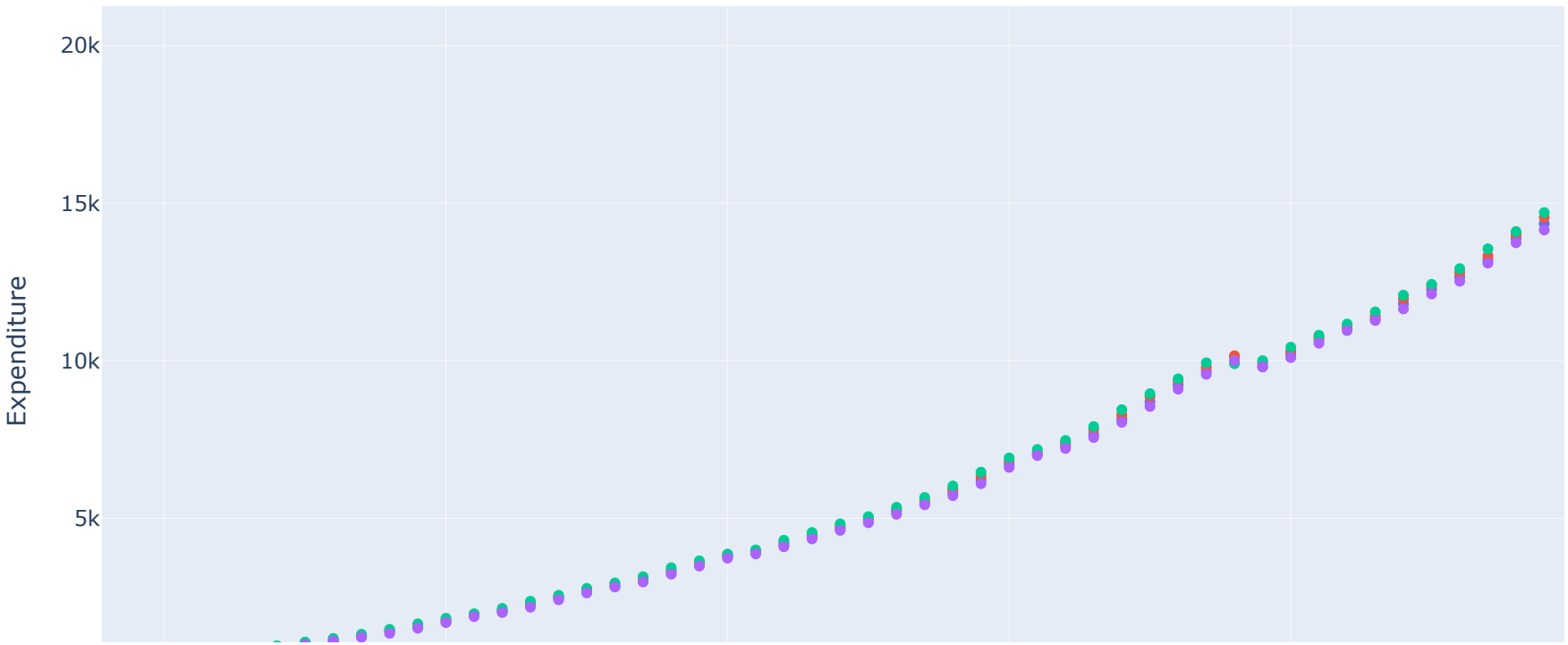
In [25]:
```
px.scatter(fl,x='Year',y='Saving Rate', color='Quarter')
```



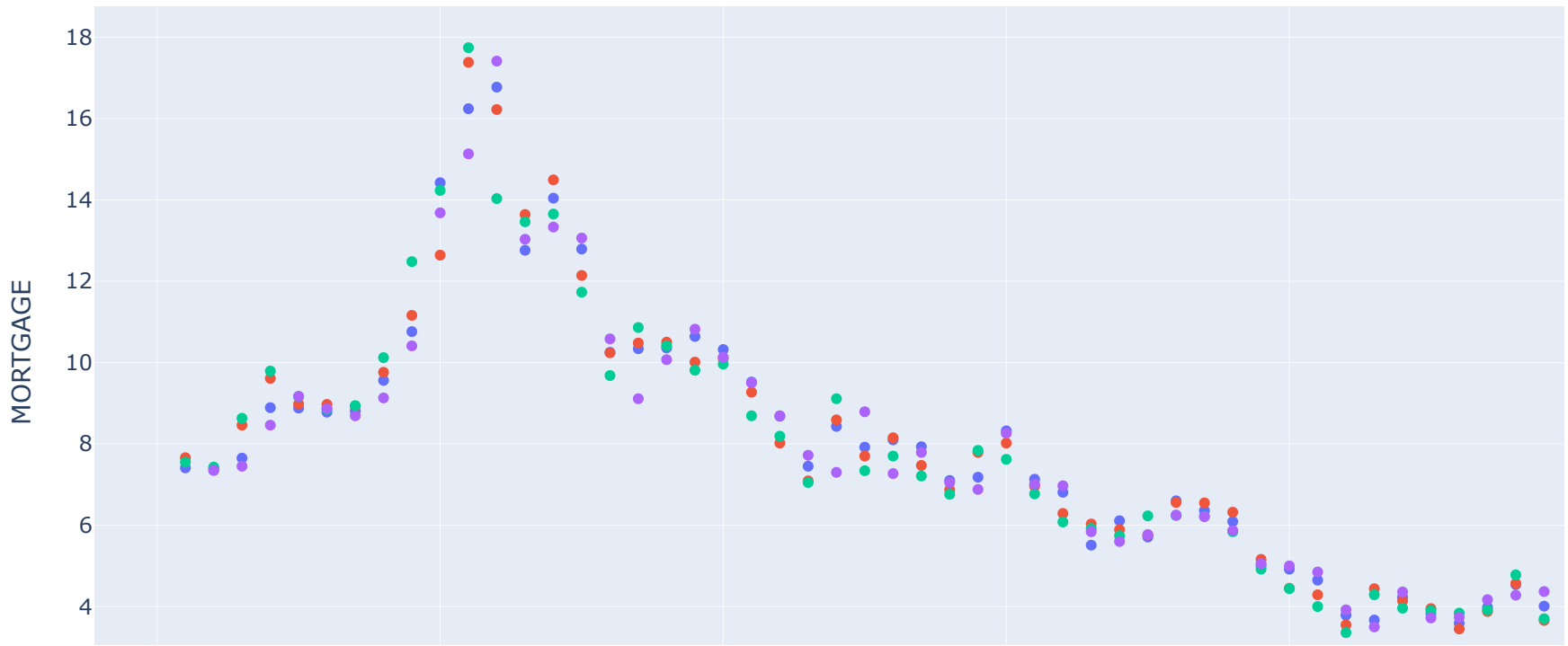
In [26]:
```
px.scatter(fl,x='Year',y='Home prices', color='Quarter')
```



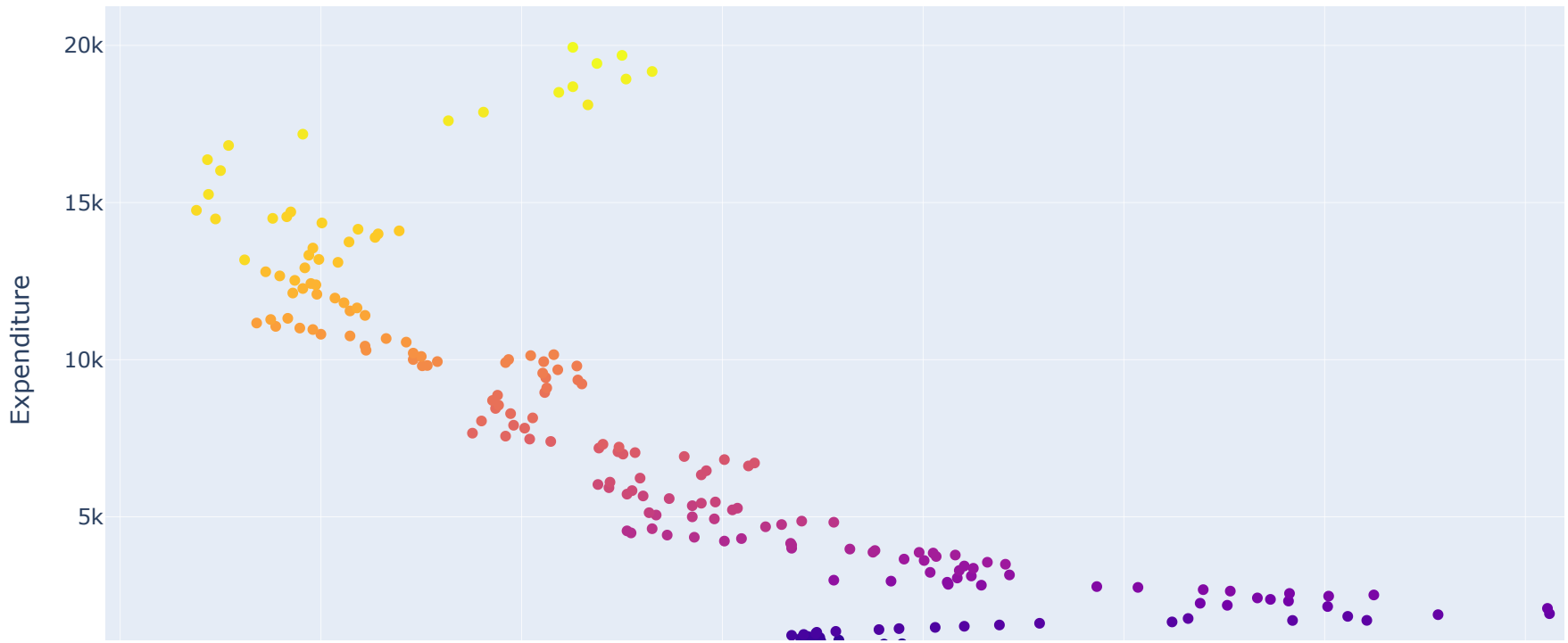
In [27]:
```
px.scatter(fl,x='Year',y='Expenditure', color='Quarter')
```



In [28]:
```
px.scatter(fl,x='Year',y='MORTGAGE', color='Quarter')
```

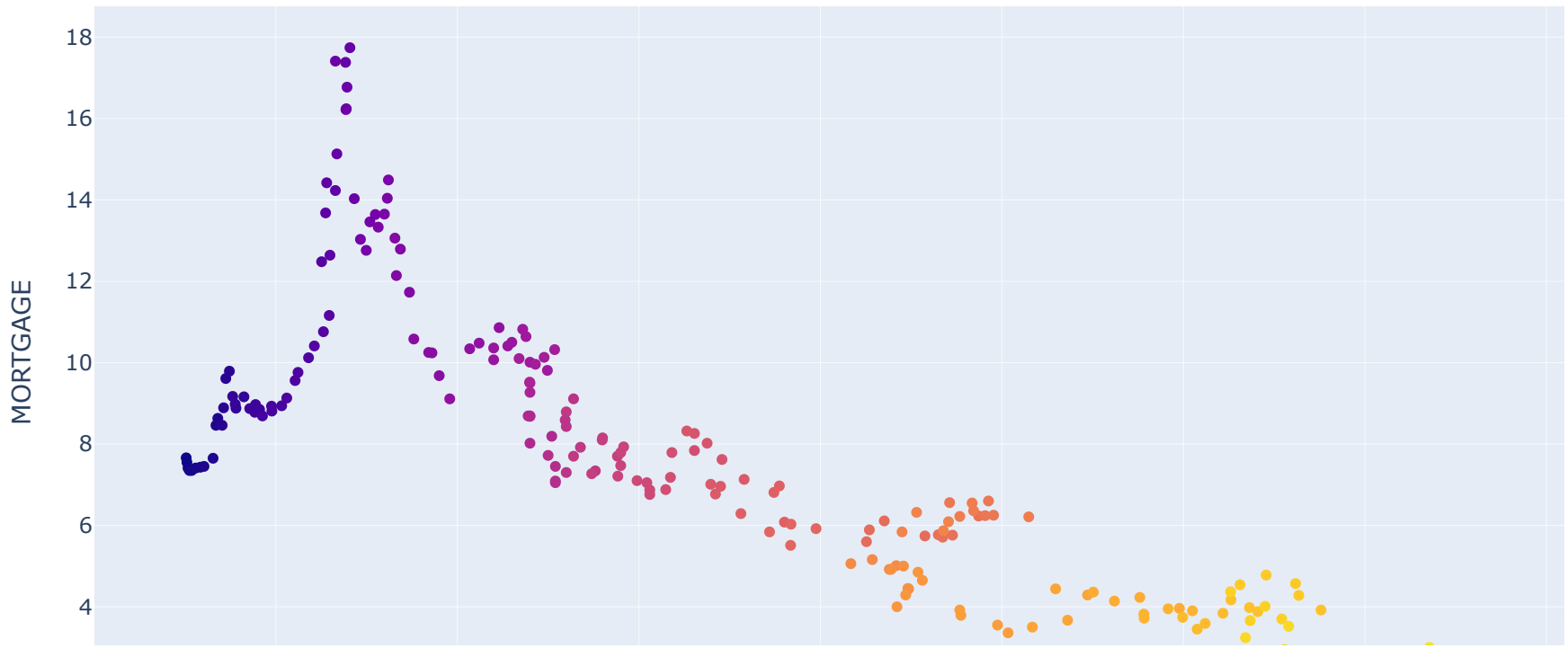


In [29]:
```
fig_1 = px.scatter(fl,x='MORTGAGE', y='Expenditure',color='Year')
fig_1
```

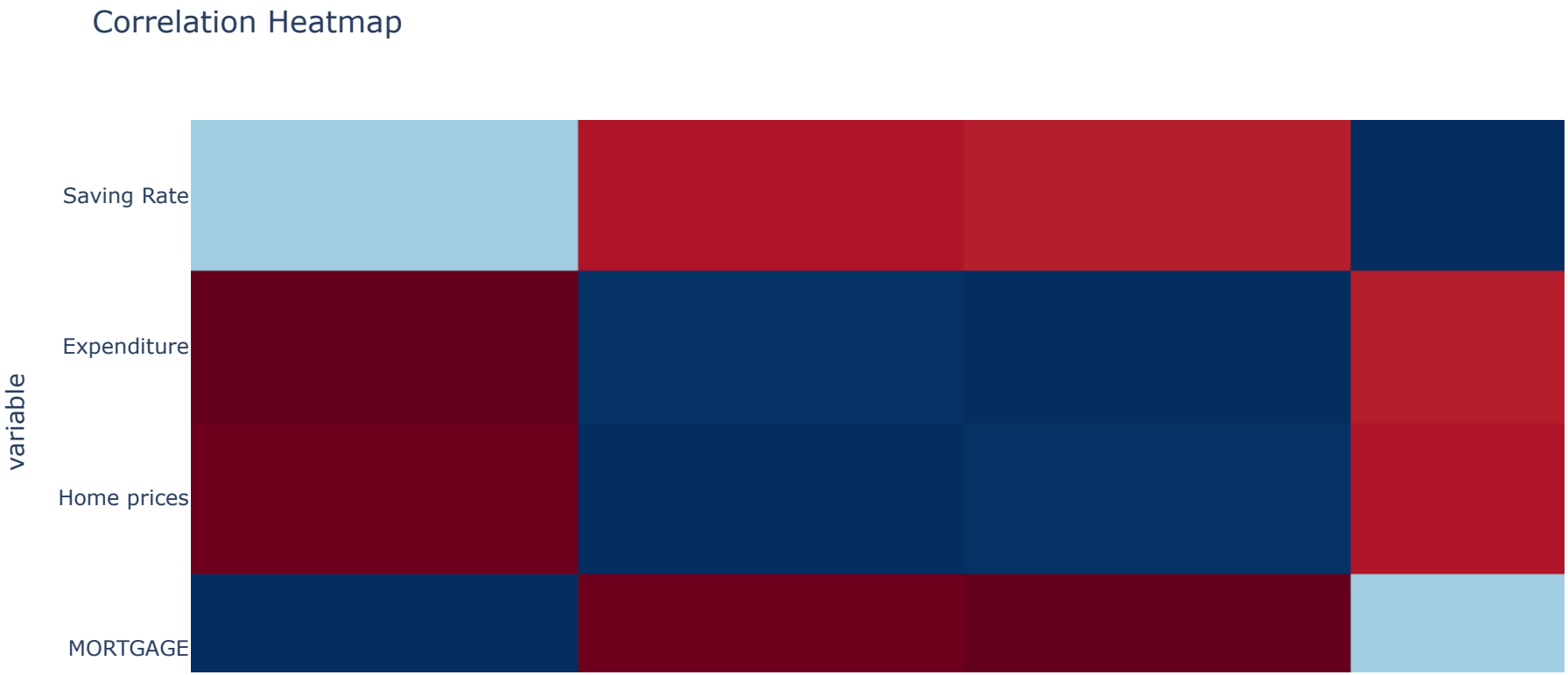


This graph tells us that morgage rates increased from 1971 to 1981, but soon started decreasing with time. This may be due to higher home prices. This cycle seemed to reverse after 2020, as mortgage rates Increased with time. According to the graph, there seems to be a negative relationship between mortgage rates and expenditure. Therefore lower mortgage rates have generally led to higher expenditure.

In [30]:
```
fig_2 = px.scatter(fl,x='Home prices', y='MORTGAGE',color='Year')
fig_2
```



In [31]:
```
#category_mean = fl.groupby('Quarter')['Home prices'].median()
#category_mean.head()
```

In [32]:
```
data = fl.loc[:,'MORTGAGE':'Saving Rate']
correlation = data.corr()
mat = correlation.reset_index().melt(id_vars='index')
fig_3 = px.density_heatmap(mat, x='index',y='variable',z='value',
```

```
                    color_continuous_scale='rdbu', title='Correlation Heatmap')
fig_3.update_layout(xaxis_title = 'Variable')
```